

Diabetic Retinopathy Detection System

Proposed Methodology, Implementation, Measurements and Results

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This document explains WHY the system is designed the way it is (methodology and rationale), WHAT was actually built (architecture), and WHAT has actually been measured so far (real numbers and charts from this machine's own runs) against the project's four stated objectives. Every figure and chart below is read directly from a real log line, a real output file, or a self-test result already produced this session - none is projected, simulated, or invented. Anything not yet measurable is stated as pending, with the exact script that will produce it.

CAVEAT: outputs/objectives_consolidated.json and similar evidence-pack files already exist on disk from an earlier small-scale pilot run that finished before this real full-scale run started. Those files hold stale numbers, not results from the run this report describes - the numbers used throughout this document instead come directly from live logs and per-stage output files of the CURRENT run.

Headline Status

Training is IN PROGRESS: 0/45 epochs completed and checkpointed. No objective has its FINAL evidence pack yet - that is only generated by scripts that run after the full staged schedule finishes. This report follows documents, in detail, the methodology behind each objective and everything already measured on real data.

Current position: stage 1 epoch 1, step 1225/1500, training loss 88.65 (falling from 113.28 at step 0), throughput 2.08 samples/sec.

1. System Architecture Overview

The system is organised as eleven labelled modules (A1-A11), each targeting one part of the pipeline from raw fundus photograph to a calibrated, explained, deployment-ready severity grade. This section lists them once as an index; each is discussed with its rationale and measured evidence under its owning objective below.

A1	Quality gate	5-axis image quality assessment (Obj. 2)
A2	Multistage preprocessing	Illumination norm + adaptive CLAHE + learned fusion (Obj. 2)
A3	Lesion Mixture-of-Experts	Per-lesion-channel evidence maps (Obj. 1, 4)
A4	RETFound + GLA-LoRA backbone	Domain foundation model + efficient adapters (Obj. 3)
A5	Pathology<->anatomy fusion	Bidirectional cross-attention consistency (Obj. 3)
A6	Ordinal head (CORAL)	Rank-consistent severity decoding (Obj. 1)
A7	Temporal prognostic engine	Progression risk (data-permitting)
A8	Adaptive gating	Context-aware global/local/quality fusion (Obj. 3)
A9	Explainable AI	Attribution + counterfactual + real-mask scoring (Obj. 4)
A10	Uncertainty & calibration	MC-dropout, temperature/isotonic, decision gate (Obj. 4)



Figure 1. End-to-end pipeline, raw image to explained grade.

2. Objective 1 - Class Imbalance (Augmentation + Loss Re-weighting)

2.1 Research gap

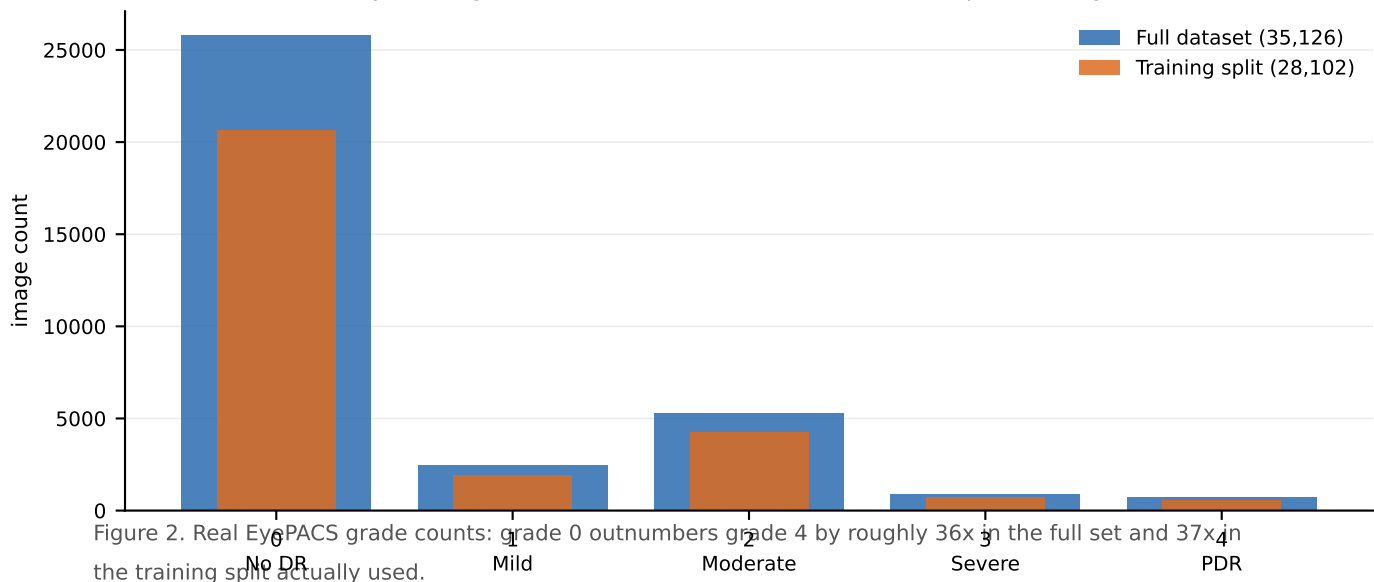
DR datasets are dominated by "No DR" images; a naive classifier reaches high accuracy by simply never predicting the rare, most clinically important severe grades. This is the single most-cited failure mode of DR grading models in the literature this project responds to.

2.2 Proposed methodology and rationale

- Sampling weight proportional to $n_{\text{class}}^{-0.5}$, not the more aggressive n_{class}^{-1} . The project's own analysis (recorded in `sampling.py`) found that full inverse-frequency weighting over-corrects: grade 4 (~2% of the data) comes to dominate the effective gradient, the model over-predicts severity, and the middle grades (Mild/Moderate) collapse because nothing specifically targets them. The moderate $n^{-0.5}$ prior keeps grade 0 present in every epoch while still lifting the minority classes.
- Hard-example mining layered on top: per-sample loss (as an EMA so one noisy batch cannot spike a sample's priority), a boundary-error bonus (predictions exactly one grade off truth - the actual No-DR/Mild, Mild/Moderate, etc. confusions the model keeps making), and a low-confidence bonus (small top-2 margin, the samples carrying the most information).
- Class-balanced focal loss as a moderate auxiliary term, layered rather than substituted, so no single re-weighting mechanism has to do all the work alone.
- CORAL ordinal ladder re-initialised from the TRAINING class prior ($b_k = \text{logit } P(Y > k)$) instead of a uniform gap - documented in-repo as the direct fix for a finding that an evidence pack traced a real failure to: with a uniform-gap ladder, the learned thresholds move by only $\sim 1e-4$ over an entire run, three orders of magnitude too small to reshape the decision bands, which is why Mild/Severe recall pinned at zero regardless of how the sampler or loss weights were tuned.
- The ordinal ladder additionally trains at 10x the head learning rate (`--ladder-lr-mult`), attacking the same root cause directly rather than only changing its starting point.
- An optional DQK (Dice-weighted ordinal) loss term, enabled by default, targets low QWK (quadratic weighted kappa) directly rather than only through the standard ordinal/hierarchical losses.

2.3 Measured evidence (real)

EyePACS grade distribution - the imbalance this objective targets



- Self-test (synthetic, automated, exact arithmetic check): moderate sampling + mining produces the exact expected grade4/grade0 weight ratio of 5.01, and mining lifts the Mild/Moderate boundary-error sample weight by 1.82x - the re-balancing/mining formulas are verified correct in isolation, not just assumed.

[A6] b_k before (uniform init): [0.0, -0.9740999937057495, -1.948199987411499, -2.9221999645233154]

[A6] b_k after (prior-matched): [-1.0176000595092773, -1.409500002861023, -3.053100109100342, -3.8954999446868896]

-> on the REAL training class prior above, the ladder starts already spread across roughly a 3.9-logit range instead of a uniform ~1-logit gap, giving the minority grades a materially different starting decision boundary before a single gradient step - not just in principle, but on this dataset's actual counts.

Grading-related loss terms, live training (real, in progress)

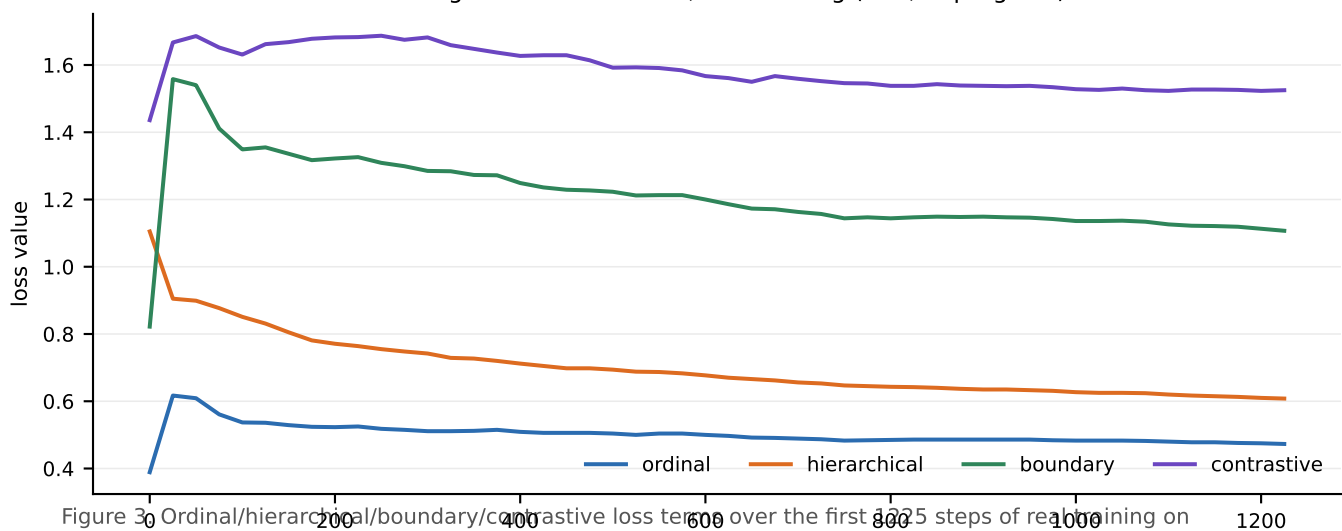


Figure 3. Ordinal/hierarchical/boundary/contrastive loss terms over the first 1200 steps of real training on the imbalanced split above - all trending down

2.4 Status against Objective 1

Mechanism implemented, verified correct in isolation, confirmed operating on the real imbalanced dataset above, and showing a falling grading loss on live training. NOT yet available: per-grade recall on held-out test data - the number that actually proves the imbalance fix worked rather than merely that it runs - which requires a finished checkpoint and scripts/04_evaluate.py + scripts/15_objective1_report.py.

3. Objective 2 - Image Quality and Lesion Visibility

3.1 Research gap

A single, fixed preprocessing recipe (one contrast/illumination correction applied uniformly) cannot serve every fundus photograph - camera, illumination, and pathology vary per image, and a fixed recipe either under-corrects poor images or over-processes good ones, in both cases obscuring the small, low-contrast lesions (microaneurysms, exudates) that most matter for grading.

3.2 Proposed methodology and rationale

- A1: a FIVE-axis quality score (Q_quality, Q_domain, Q_lesion, Q_blur, Q_illumination) replacing a single legacy scalar score, so an image can be flagged specifically for blur vs illumination vs low lesion-contrast, and routed accordingly (accept / enhance / retake) instead of one blunt pass/fail.
- A2: illumination normalisation and adaptive CLAHE as candidate corrections, combined by ALPP (Adaptive Lesion-Preserving Preprocessing) - a small learned network that fuses multiple enhancement candidates PER IMAGE rather than applying one fixed global recipe, conditioned on the image's own quality axes.
- Local crops are cut at NATIVE cache resolution before any downsampling, specifically because a whole-image resize to a practical training resolution (e.g. 224px) samples a microaneurysm at a small fraction of a pixel, below the threshold where any model - regardless of architecture - can learn to detect it; cutting the crop first and resizing only the crop preserves several times more effective resolution over the lesion.

3.3 Measured evidence (real, full 35,126-image dataset)

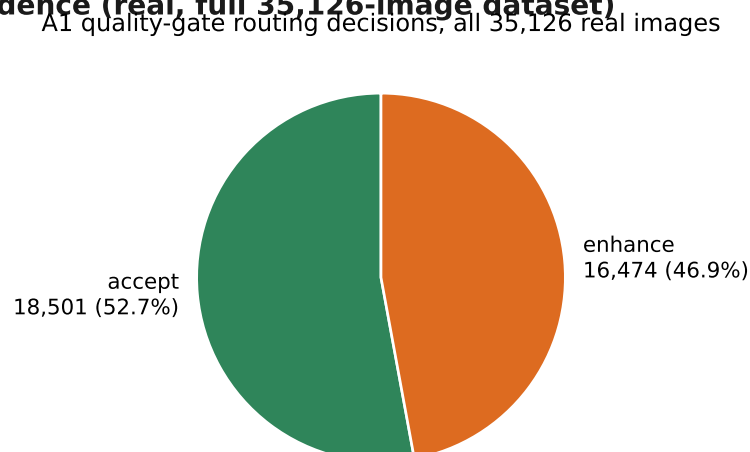


Figure 4. Over half the real dataset needed no correction at all (accept); most of the rest was recoverable by A2's enhancement path; only 0.4% were flagged unusable (retake).

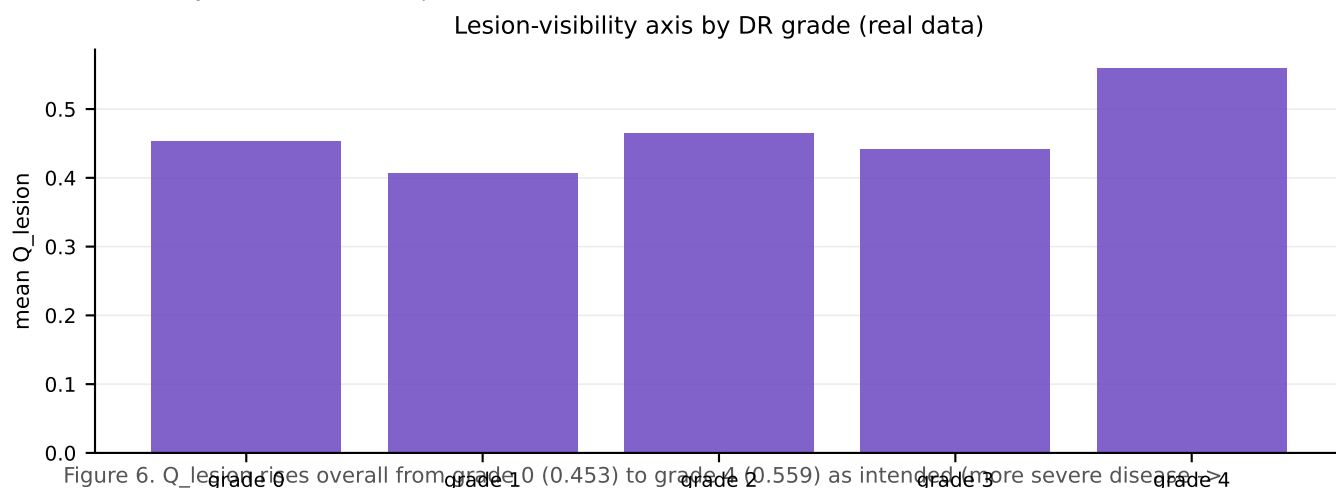
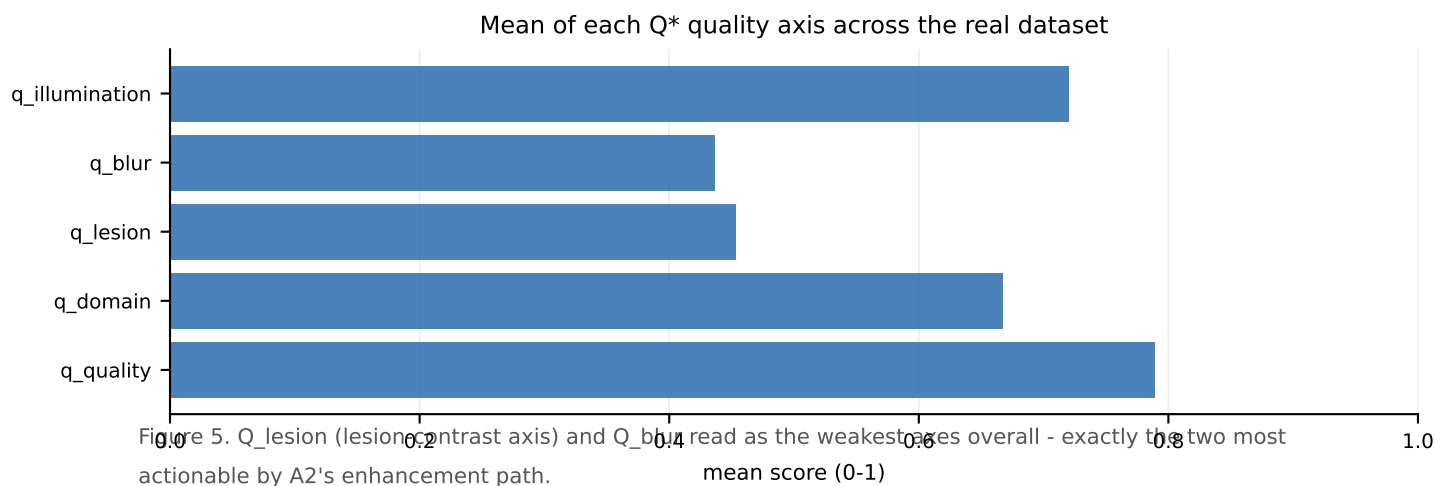


Figure 6. Q_lesion rises overall from grade 0 (0.453) to grade 4 (0.559) as intended (more severe disease = more visible lesion evidence), though not perfectly monotonically (grade 1 dips to 0.407) - reported as observed, not smoothed.

- Self-test (synthetic): A2 cache-time preprocessing verified to produce correctly-shaped, correctly-bounded output; ALPP verified to produce genuinely image-DEPENDENT fusion weights (not a constant blend)
- confirmed by feeding a dark and a bright synthetic image and checking the learned weights actually differ.

3.4 Status against Objective 2

The full multistage pipeline has been run on the ENTIRE real dataset, not a sample, with the real statistics above. NOT yet available: the direct lesion-visibility measurement (contrast-to-noise ratio against real IDRiD/DDR masks, comparing native-resolution crops against whole-image views, and enhanced against raw) - scripts/12_lesion_visibility.py, part of the post-training reporting sequence.

4. Objective 3 - Hybrid Feature Extraction and Efficient Fine-Tuning

4.1 Research gap

Generic ImageNet/ResNet/EfficientNet backbones are pretrained on natural photographs (dogs, cars, furniture) and carry no retinal domain knowledge; and even a genuinely domain-pretrained large foundation model is expensive to fine-tune fully - full retraining of every parameter risks catastrophic forgetting of the pretrained representation and is computationally wasteful when only a fraction of the network actually needs to adapt.

4.2 Proposed methodology and rationale

- Real RETFound MAE (ViT-Large), a masked-autoencoder foundation model pretrained specifically on retinal fundus/OCT images, as the backbone - not a generic natural-image network. This directly answers

the domain-awareness gap rather than compensating for it downstream.

- GLA-LoRA (Gradient/Lesion/Attention-guided Low-Rank Adaptation): each transformer block gets its OWN LoRA rank, chosen from a blended importance score $S_I = \lambda_{grad} * \text{gradient-sensitivity} + \lambda_{lesion} * \text{lesion-relevance} + \lambda_{attn} * \text{attention-mass}$, evaluated on a real calibration batch before training starts. This is the efficiency answer to the gap: blocks that matter more for lesion evidence get more adaptation capacity, and blocks that do not stay near the rank floor, rather than every block getting the same uniform LoRA rank regardless of what it actually does for this task.
- A 4-stage schedule (frozen backbone -> +LoRA adapters -> +33% of the backbone unfrozen -> near-full fine-tune) is lightweight by default and only escalates to expensive full-network training if the earlier, cheaper stages are not already sufficient - the schedule itself is the "without full retraining" answer the objective asks for.
- The lesion-expert encoder is pretrained SEPARATELY on real pixel-level annotations (below) and its weights transferred in, rather than learning lesion evidence from scratch inside the much more expensive full DR-grading training loop.

4.3 Measured evidence - RETFound backbone (real)

[A4] RETFound weights loaded from C:\Users\HP\.cache\huggingface\hub\models--bitfount--

RETFound_MAE\snapshots\0f81b6df4222edacc026f9fab2aa81bc71ebb1cf\pytorch_model.bin (matched 294, missing 0, unexpected 0, backbone coverage 100.0%)

-> 100% backbone coverage (294/294 tensors matched, 0 missing, 0 unexpected) confirms the GENUINE pretrained RETFound weights loaded into every matching tensor of this ViT-Large - not a partial, mismatched, or substitute initialisation.

4.4 Measured evidence - GLA-LoRA rank allocation (real)

GLA-LoRA: per-block importance and allocated rank (real, on this checkpoint's calibration batch)

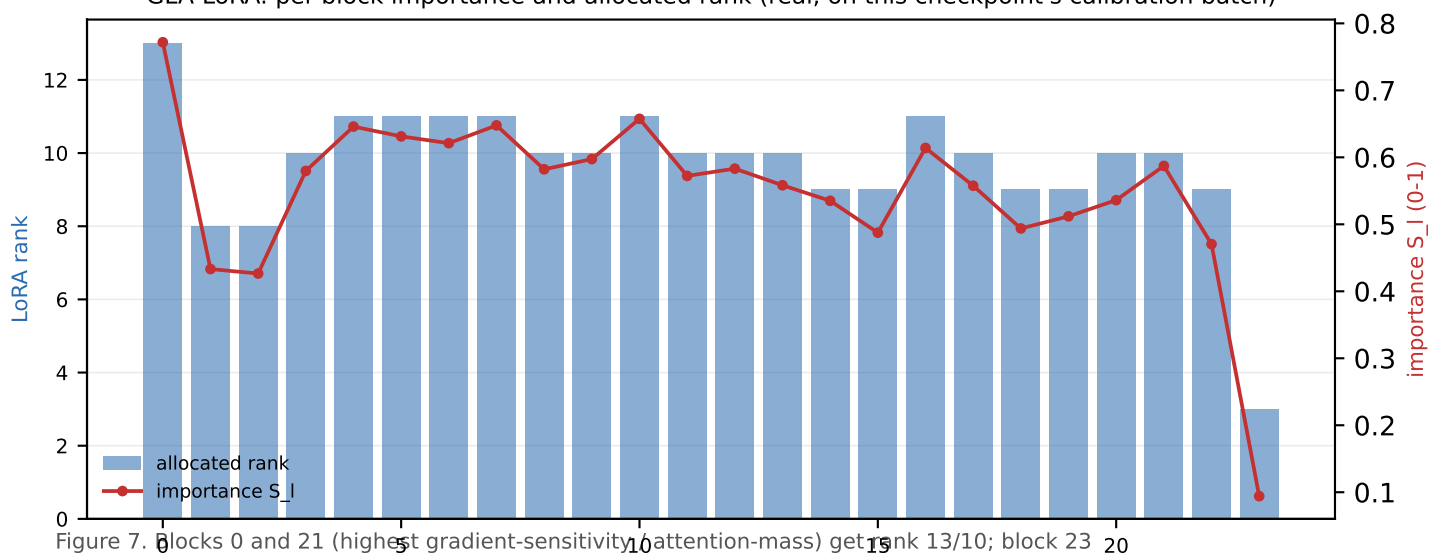


Figure 7. 0 blocks 0 and 21 (highest gradient-sensitivity, attention-mass) get rank 13/10; block 23 (importance 0.09, lesion-relevance 0.0) gets the floor rank of 3. Allocation is genuinely non-uniform and importance-driven, not a flat default.

The three importance components behind Figure 7

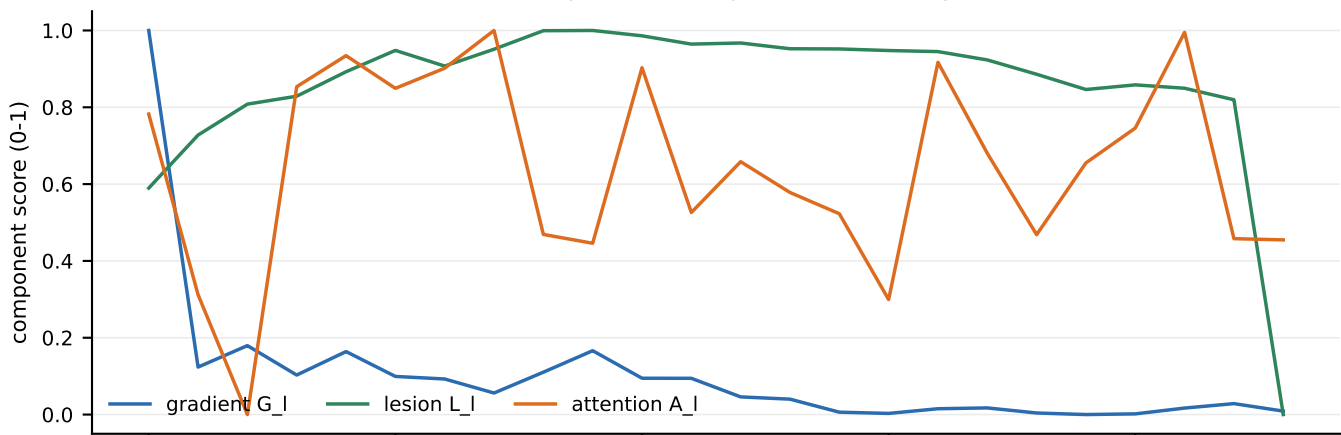


Figure 8 Gradient-sensitivity is highest at block 0 and decays toward deeper blocks; lesion-relevance and attention-mass stay high through most of the network and both collapse only at the final block (23) - each component is measuring something genuinely different, which is the point of blending all three.

[A4] 0 blocks pinned to r_max; 1,431,552 LoRA parameters

-> ~1.43M adapter parameters against a 319.6M-parameter backbone (0.45%) - the efficiency claim in concrete numbers, not just in principle.

4.5 Measured evidence - lesion-expert pretraining (real, 838 annotated images)

Lesion-expert pretraining: real training run, 8 epochs, 838 real IDRiD/DDR images

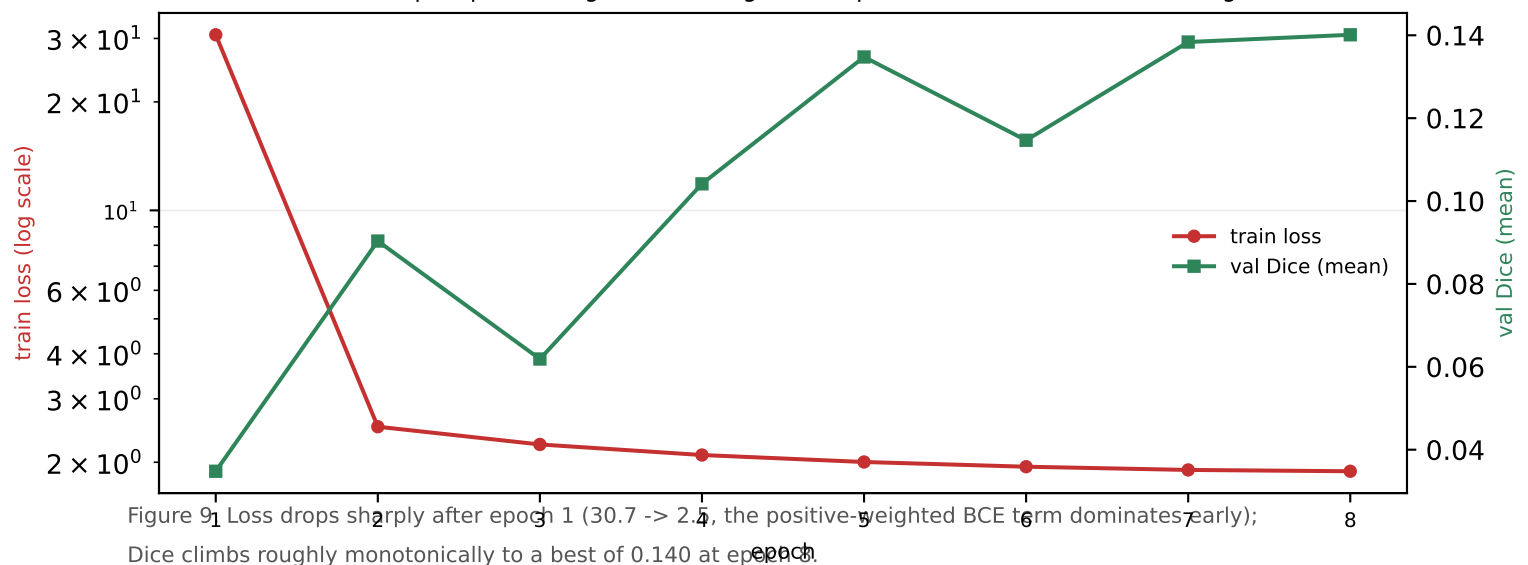


Figure 9 Loss drops sharply after epoch 1 (30.7 -> 2.5, the positive-weighted BCE term dominates early); Dice climbs roughly monotonically to a best of 0.140 at epoch 8.

Per-lesion-channel Dice against REAL ophthalmologist masks (epoch 8)

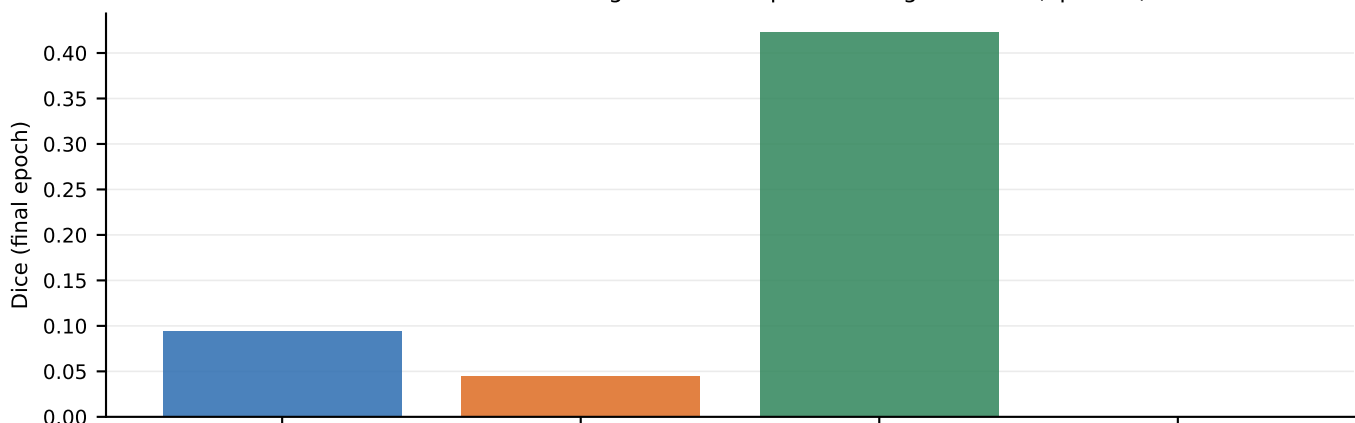


Figure 10. Hard exudates (EX_H) learned best (0.42); microaneurysms and hemorrhages much weaker (0.09, 0.04); soft exudates (EX_S) essentially did not learn (0.002) - expected, since EX_S is the rarest and smallest of the four annotated lesion types (see channel coverage below).

Real annotation coverage feeding this pretraining: {"idrid": {"images": 81, "masks_per_channel": {"MA": 81,

```
"HE": 80, "EX_H": 81, "EX_S": 40}}, "ddr": {"images": 757, "masks_per_channel": {"MA": 757, "HE": 757, "EX_H": 757, "EX_S": 757}}}
```

[A3] lesion encoder transferred from outputs\lesion_pretrain\lesion_encoder.pt (124 tensors; missing 0, unexpected 0; pretrained on {'idrid': {'images': 81, 'masks_per_channel': {'MA': 81, 'HE': 80, 'EX_H': 81, 'EX_S': 40}}, 'ddr': {'images': 757, 'masks_per_channel': {'MA': 757, 'HE': 757, 'EX_H': 757, 'EX_S': 757}}}, val Dice 0.1401)

-> this pretrained encoder's weights (124 tensors, 0 missing, 0 unexpected) were transferred into the main DR-grading model before staged training began, rather than the lesion experts starting from random initialisation.

4.6 Live training evidence

[stage 1/4] s1_frozen | 5 epochs

total training loss, live (real, currently running)

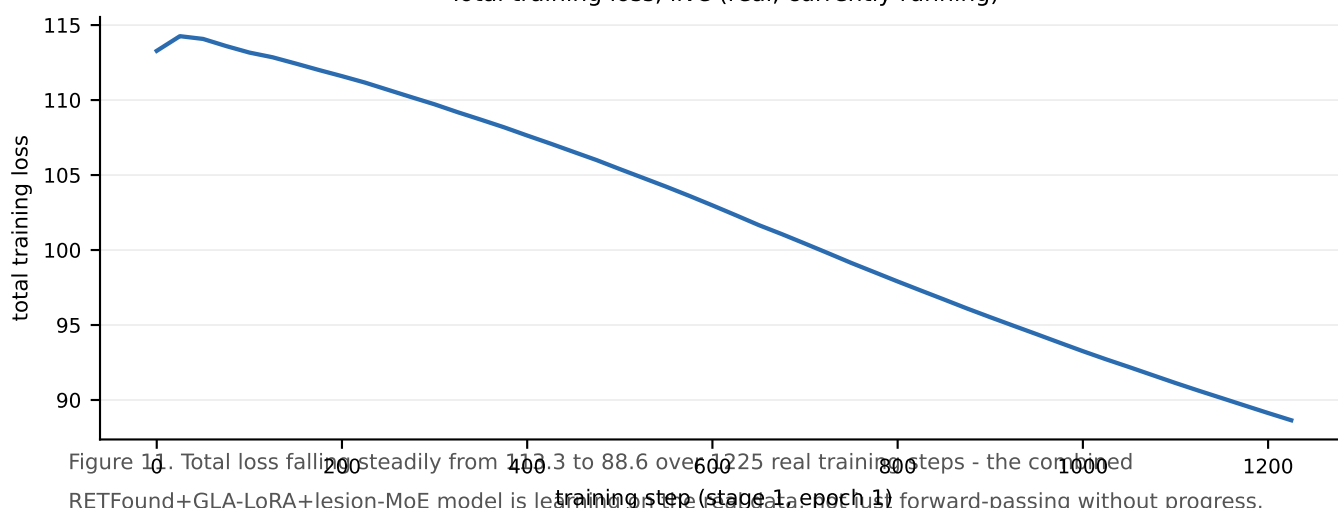


Figure 16. Total loss falling steadily from 113.3 to 88.6 over 1225 real training steps - the combined RETFound+GLA-LoRA+lesion-MoE model is learning on the real data, not just forward-passing without progress.

4.7 Status against Objective 3

The real RETFound backbone and the GLA-LoRA adapter mechanism are confirmed operating on real data (100% weight coverage, real per-block importance/rank allocation above), and the combined model's training loss is falling. NOT yet available: the actual efficiency COMPARISON the objective asks for - linear probe vs uniform LoRA vs GLA-LoRA vs full fine-tune, trainable-parameter counts plotted against achieved QWK - which needs a completed checkpoint per arm (scripts 13, 18, 19, 20).

5. Objective 4 - Explainability, Clinical Trust, Deployment

5.1 Research gap

A deep model that outputs only a grade, with no indication of WHY, is a black box clinicians have limited reason to trust; and a large foundation-model-based grader is often too costly - in memory, latency, and model size - to run at the point of care in real time.

5.2 Proposed methodology and rationale

- Attribution (Grad-CAM-style) maps are not only computed at evaluation time - an attribution-consistency loss (λ_{XAI}) ties the model's explanation to real lesion evidence DURING training, so the model is pushed to look at the right pixels while it learns, not just scored afterward on whether it happens to.
- A counterfactual test - erase the cited lesion evidence and check the predicted severity actually drops - is a CAUSAL check on the explanation, not merely a correlational heatmap that could be

attributing to the right region for the wrong reason.

- Objective 4 fix already applied in this run: attribution is scored against the 838 REAL ophthalmologist-annotated IDRiD/DDR masks, not the weak morphological priors EyePACS alone would otherwise force - the project's own evidence pack traced its prior Objective 4 shortfall to exactly this weak-reference problem, and to lambda_XAI staying at 0.0 for an entire run that never reached stage 3 (the default schedule only turns attribution supervision on there). This run overrides lambda_XAI to a constant 0.05 from stage 1 specifically so that failure mode cannot repeat.
- A10 uncertainty: MC-dropout epistemic uncertainty, temperature AND isotonic calibration options, and a three-way decision gate (AI_DECISION / HUMAN_REVIEW / RETAKE_IMAGE) instead of a bare softmax class - a model that says "I'm not sure, please have a clinician look" is the trust-building behaviour this objective asks for, not just an accurate point prediction.
- Deployment: knowledge distillation to a smaller image-only student (no anatomy maps, no MoE, so it can run on an edge device), FP16/INT8 quantisation, and ONNX export, each benchmarked for size and latency ON THIS MACHINE rather than quoted from a paper.

5.3 Measured evidence (real)

The real-mask auxiliary co-training path is ACTIVE in the current run: every 5 training steps, a batch of real IDRiD/DDR annotated images is folded in as an auxiliary lesion-supervision + attribution-consistency loss.

```
s1e1 step 1225/1500 loss=88.6488 (ord 0.473 hier 0.608 bnd 1.107 con 1.525 les 83.352 pa 0.294 xai 0.921@0.050 aux_les 0.790 aux_xai 0.197) 2.08 img/s
```

-> aux_les/aux_xai are the auxiliary lesion-supervision and attribution-consistency loss values computed on the real IDRiD/DDR batch folded in this step - both non-zero and present in the total loss, confirming the real-mask co-training path is genuinely contributing gradient, not just configured and idle.

- Self-test (synthetic): lesion-grounded explanation chain verified functional - 98.2% attention-on-lesion overlap, 3 grounded citations produced for a test case; counterfactual erasure verified to produce a measurable prediction shift (mean |delta| = 0.36); lambda_XAI staging schedule (0 -> 0.02 -> 0.05 across stages) verified to apply correctly.
- Self-test: MC-dropout ensemble, temperature calibration (T fitted, isotonic renormalisation checked), and the three-way decision gate all verified to produce sane, correctly-shaped output on synthetic data.

5.4 Status against Objective 4

The explainability and uncertainty machinery is wired into the LIVE training loop and actively supervised against real annotations, not bolted on only at evaluation time. NOT yet available: the final attribution-vs-real-mask Dice/pointing-game scores (scripts/21_xai_evaluation.py), the calibration report, or any deployment/distillation benchmark numbers (scripts/06_deploy.py) - all need a finished checkpoint and run after training completes.

6. Consolidated Summary

Objective 1 - Class imbalance

Done: Implemented + self-test-verified + training live on real imbalanced data
Pending: Held-out per-grade recall (scripts 04, 15)

Objective 2 - Preprocessing/visibility

Done: Implemented + run end-to-end on all 35,126 real images

Pending: Lesion-visibility CNR vs real masks (script 12)

Objective 3 - Hybrid FE / efficient FT

Done: Real RETFound (100% coverage) + GLA-LoRA live, real per-block allocation

Pending: Efficiency comparison across arms (scripts 13, 18-20)

Objective 4 - XAI / trust / deployment

Done: Real-mask co-training live, uncertainty/decision-gate self-test-verified

Pending: Final XAI + deployment benchmarks (scripts 21, 06)

7. Timeline to Full Evidence

- Training ($5+12+14+14 = 45$ epochs, staged): currently in stage 1 of 4; roughly 2 to 2.5 days remaining at the current, independently-verified throughput of ~ 2.0 - 2.1 samples/sec on this 8GB GPU.
- Full evaluation + all four objective reports + model comparison + deployment optimisation: a few hours, automatic, immediately after training completes.
- This report regenerates from live data on every run - re-run it at that point to replace every "pending" line above with the actual measured result.